



Manav Marmat

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B. Tech. in Electronics and Communication Engineering

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GitHub Profile

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EDUCATION

- Maulana Azad National Institute of Technology (MANIT), Bhopal, India** July 2023 – Present
B.Tech. Electronics and Communication Engineering CGPA: 8.55/10.00

EXPERIENCE

- Computer Science and Automation Department, Indian Institute of Science** May 2025 – July 2025
Indian Academy of Sciences Summer Research Fellow Bengaluru
 - Created a **WebGL2 based visualization tool** for contour trees. Enabled scalar function analysis through branch selection.
 - Used **impostor geometry** for optimized rendering.
 - Technologies used: JavaScript, WebGL2, GLSL ES 3.00, HTML5, CSS3.
- "BisKit" – BIS (Bureau of Indian Standards) Hackathon Project, MANIT** Dec 2024 – Jan 2025
Team Leader and Lead Game Developer Bhopal
 - Used Stable Diffusion 2 for GenAI based asset generation. Tracked and resolved gameplay and integration bugs through iterative testing, resulting in a cash prize of INR 2,00,000.
 - Technologies used: JavaScript, p5.js, Python3, HTML5, CSS3.

PERSONAL PROJECTS

- PReSPIDAR - Pulsar-Polarisation REsolved Single Pulse Interactive Data AnalyseR** Dec 2025 – Jan 2026
An open-source scientific web-app for the interactive visualization of astronomical observations of pulsars.
 - Deployed on a academic Linux server using Gunicorn with Uvicorn workers for concurrency. Created a reverse-proxy that routes the backend API as direct configuration of Nginx could not be used, and added lazy loading for plots, minimizing the initial load time of plots by rendering them only when they came into view.
 - Technologies used: React, Plotly.js, JavaScript, Python, FastAPI, Gunicorn, Uvicorn, Node.js, Postman, Linux.
- AudioPrism - An AI web-app for audio stem separation (vocals and instruments)** Oct 2025 – Dec 2025
A UNet-CNN based model that segments STFT spectrograms of audio into constituents through trained masks
 - Trained the model on MedleyDB2.0 dataset using an NVIDIA RTX 6000 GPU in a HPC environment, with PBS job scheduling and checkpointing for long running training. Deployed the inference pipeline as a full-stack application using FastAPI and React, hosted across Azure, Render, and Vercel.
 - Technologies used: Python, PyTorch, CUDA, UNet, STFT, FastAPI, React, Node.js, Azure, Vercel, Render.
- Binary Star Simulation in p5.js** Nov 2024
A 2D simulation using OOPs to simulate orbital mechanics, radial velocity, and composite light curves of stellar binaries.
 - Optimized rendering performance by controlling canvas pixel density, reducing rendering overhead on high-DPI displays.
 - Technologies used: JavaScript, p5.js, HTML5, CSS3.
- BhaataPhod – A 2D bullet-hell style Space Shooter Game | Released on itch.io** Nov 2024
Made with C++ using SFML and object-oriented principles with smart pointer managed game entities and custom sprites.
 - Technologies used: C++, SFML, Piskel.

TECHNICAL SKILLS

Technologies and Languages: C++, C, Javascript, Python, HTML, CSS

Developer Tools: Git/Github, VS Code, Visual Studio, Postman, Google Colab

Frameworks and Libraries: WebGL2, OpenGL GLUT, p5.js, Ren'Py, SFML, PyTorch, React, Node.js, Matplotlib, Plotly.

Cloud Platforms: Azure, Render, Vercel, MongoDB Atlas.

Soft Skills: Effective Communication, Design Thinking, Team Resource Planning and Leadership.

CS Coursework: Data Structures and Algorithms (A+), Computer Networks (A+), Operating Systems, OOP.

ECE Coursework: Microprocessor and Microcontrollers Lab (A+), Digital Signal Processing Lab (A), Linear ICs (A), Information Theory and Coding (A), Digital Image Processing (A).

Electronics Tools: LTSpice, AMD Xilinx Vivado, TINA-TI, Keil uVision5, Cadence Virtuoso, KiCAD, Arduino IDE.

ACHIEVEMENTS

- Centre for Nano Science and Engineering Summer School, IISc. Bengaluru** Jun 2026
Completed a certified workshop on recent advances in semiconductor fabrication
- Graduate Aptitude Test in Engineering (GATE) 2026** Feb 2026
AIR 5917 in EC (Score: 471/1000) and AIR 7170 in CS (Score: 506/1000)
- CS50x - Introduction to Computer Science** Jul 2024
Credential ID - 22244739-4a74-41de-a5f4-ccff1cd66548

POSITIONS OF RESPONSIBILITY

- Coordinator, BIS (Bureau of Indian Standards) Club, MANIT Bhopal** Sep 2024 – Present
Led team to victory at BIS Hackathon 2024, coordinated event planning and management for the club.